

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12409793
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Tansingco; Edward

Protective shell for passenger vehicle door

Abstract

A protective shell for protecting an interior door panel of a vehicle has an inner face with a protruding door pull defining a door well for receiving a user's fingers and a door sill leading to a window opening. A deformable sheet of hard plastic with an exposed side and a reverse side is profiled to fit over the inner face of the vehicle door interior panel. An outwardly directed top surface portion for rests on the door sill. A protrusion on the exposed side of the deformable sheet for encloses the vehicle door pull. The protrusion has a locking well responsive to downward pressure to fit snugly into the door well to hold the protective shell securely in position on the interior door panel. A separate flange component has a main surface and a downward flange at its upper extremity tuckable over a top edge of the door sill. The main surface is attachable to the outwardly directed top surface portion.

Inventors:	Tansingco; Edward (Santa Ana, CA)
Applicant:	Tansingco; Edward (Santa Ana, CA)
Family ID:	83006909
Appl. No.:	17/748391
Filed:	May 19, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220274538 A1	Sep. 01, 2022

Related U.S. Application Data

continuation-in-part parent-doc US 16914649 20200629 US 11358540 child-doc US 17748391

Publication Classification

Int. Cl.: B60R13/02 (20060101)

U.S. Cl.:

CPC **B60R13/0243** (20130101); **B60R13/0206** (20130101); B60R2013/0293 (20130101)

Field of Classification Search

CPC: B60R (13/0243)

USPC: 296/146.7; 296/153

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2600296	12/1951	Jacobs	N/A	N/A
3855898	12/1973	McDonald	N/A	N/A
5811719	12/1997	Madden, Jr.	296/146.7	F41H 5/226
6248200	12/2000	Dailey	264/250	B29C 44/145
6926341	12/2004	Addesso et al.	N/A	N/A
8615932	12/2012	Setina	N/A	N/A
2006/0261633	12/2005	Winborn	N/A	N/A
2009/0151827	12/2008	Thompson	N/A	N/A
2013/0055643	12/2012	Setina	N/A	N/A
2014/0054914	12/2013	Teoh et al.	N/A	N/A
2015/0097392	12/2014	Locke	N/A	N/A
2020/0207191	12/2019	Mullen	N/A	B60R 13/0243

Primary Examiner: Pape; Joseph D.

Attorney, Agent or Firm: LAUBSCHER & FRETWELL, P.C.

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation-in-part of co-pending U.S. application Ser. No. 16/914,649, filed Jun. 29, 2020, the contents of which are herein incorporated by reference.

FIELD OF THE INVENTION

(1) This invention relates to the field of vehicle accessories, and more particularly to a protective shell for protecting the door panel of a passenger vehicle, and a method of applying the protective shell.

BACKGROUND OF THE INVENTION

(2) The average price of new cars has been trending upward and most car manufacturers have been providing new models with more plush interiors in response to consumer demand. Most passenger cars sold today have interiors lined with easily damaged surfaces made of highly finished plastic, painted plastic, cloth, vinyl, leather, skinned-foam, finely finished wood trim and other soft upholstery materials serving primarily cosmetic, non-structural purposes. The interior door panels

are designed primarily to be aesthetically pleasing and may serve to provide places to mount door handles, lock buttons, window controls, door pulls, arm rests, cup holders, speakers and small storage compartments. There are many day to day situations during which a car's interior panels are exposed to cosmetic damage by its occupants or from contact with other items that are temporarily placed on the seat of the vehicle. Items which are regularly loaded onto car seats such as child car seats, sporting equipment, groceries such as canned goods and glass bottles, paper or plastic boxes, tools, small pieces of furniture, luggage or other equipment can jostle around during loading or transport and cause scuffs, scrapes and tears on the panel. Other potential causes of damage to interior panels come from the occupants themselves who can easily spill liquids onto the panels, abrade the plastic and upholstery with worn jewelry. Another potential cause of damage is from contact with pet claws, teeth and saliva when owners travel with their dogs who often try to look out the window while placing their paws on the arm rest and sill.

(3) Damaged panels are very costly to repair or replace, diminish the enjoyment of the owner who paid such a high price for their car, and ultimately reduce the resale value of the vehicle. Interior door protectors are described in prior art but their installation is elaborate, requiring fasteners such as screws or adhesives to hold them in the intended operating position which consequently puncture or mar the interior surfaces hence diminishing the purpose of protecting the vehicle to maintain its original condition. Some interior door protectors are made of flexible materials such as cloth have been known to be blown out the window when the car is traveling at normal highway speeds thus requiring them to be held down by hook and loop strips (like Velcro) of which one side of the fastener must be affixed to the interior surface with adhesive where it cannot be tied down to some other feature of the vehicle interior. Further, they are not convenient to install or remove so that frequent removal and installation is not possible.

(4) Therefore, a car door panel protector which will protect the panel from day to day hazards, and will itself cause no damage during its installation and use, and which is easily removable without adhesives or fasteners, can preserve the appearance and value of the vehicle is needed.

(5) Many car owners take pride in their vehicles and want to protect their investment but, like most people, are also extremely busy and so would probably prefer to have a car door panel protector that they can install and remove quickly and easily such as when they are cleaning the interior of the car.

(6) U.S. Pat. No. 6,926,341 to Adesso discloses a pad which is secured to the door panel with a flap to which hang tabs are secured and inserted in the well between the interior portion of the car door and the window. The hang tabs in Adesso's apparatus are inserted in between the well and window glass, making contact with the window glass and therefore tend come out when the window is rolled up. Further, the tabs which are in contact with the glass tend to cause scratches to any tint that is applied to the glass. Due to their placement, the tabs also make contact with and cause indentations in the inner door window beltline seal which is normally affixed by the original vehicle manufacturer on the return face of the panel. This window seal is usually flocked with fine fibers where it is intended to make contact only with the window glass to keep outside air from entering the cabin and acts as a brush when the window is rolled up and down is very sensitive to damage from contact with other surfaces than the window.

(7) U.S. Pat. No. 8,615,932 to Setina discloses a barrier element for protecting an interior vehicle door panel which is comprised of a body shaped to fit over the vehicle door interior panel. The panel of Setina which is intended to be installed in law enforcement vehicles, is made from a rigid plastic sheet, such as 0.187 thermoplastic olefin (TPO), is very sturdy and will presumably provide ample protection for a car's interior door panel against the actions of a detainee within the vehicle and therefore will not be pliable. Any recesses are stated to be covered by a plane or convex surface to prevent concealment of contraband. It also requires that the panel be mounted to the door using screws or adhesive which will certainly reduce the resale value of the vehicle and will not easily allow the average consumer to frequently remove and install the apparatus.

(8) U.S. Pat. No. 3,855,898 to W McDonald discloses a series of transparent protective panels made of tough, bullet resistant plastics for windshields and windows of a vehicle and means for attaching them to a vehicle. It describes fabric straps or metal clips secured to the panels which engage with window trim means of a vehicle. FIG. 4A shows a foot attaching clip or bracket 72 and a second positioning clip or strap 73 both secured to a panel 70 by means 74. This clip and the second positioning clip 73 are separate components attached with fasteners 74 to the panel 70, are not formed unitary with the protective panel thereby increasing the cost of manufacturing. Further, the assembly of FIG. 4A will not allow unobstructed use of the door.

SUMMARY OF THE INVENTION

(9) The invention provides a protective shell for protecting a passenger car interior door panel to preserve the value and enjoyment of the vehicle comprises a thin hard plastic shell shaped to fit over the vehicle door interior panel. The protective element will allow use of the door and window as provided by the original equipment manufacturer. The protective element is removably securable to the vehicle door panel without fasteners or adhesives by interfacing with the door pull and the interior surface of the panel such that the protective element can be easily installed and removed to expose or cover up the original panel on a frequent basis.

(10) In accordance with a first aspect of the invention there is provided a protective shell for protecting an interior door panel of a vehicle, the interior door panel having an inner face with a protruding door pull defining a door well for receiving a user's fingers and a door sill leading to a window opening, the protective shell comprising: a deformable sheet of hard plastic with an exposed side and a reverse side, the reverse side being profiled to fit over the inner face of the vehicle door interior panel; an outwardly directed top surface portion for resting on the door sill; a protrusion on the exposed side of the deformable sheet for enclosing the vehicle door pull, the protrusion having an upper ledge; a locking well extending downwardly from the upper ledge and responsive to downward pressure to fit snugly into the door well to hold the protective shell securely in position on the interior door panel; and a separate flange component having a main surface and a downward flange at its upper extremity tuckable over a top edge of the door sill, said main surface being attachable to said outwardly directed top surface portion.

(11) In accordance with embodiments of the present invention installation of the portable car door protective shell requires no fasteners or adhesives.

(12) A passenger car door typically has an interior panel with a horizontally planar sill which has a short, downward return at its top extremity to the outer surface of which is affixed a flexible rubber seal, known as an inner door beltline window seal, running horizontally across the width of the portion of the window which slides up and down. The purpose of this seal is to keep outside air from entering the cabin and is lined with a flocked material where it meets the inside surface of the window to reduce friction when operating the window. The profile of the inner door beltline window seal typically forms an upward oriented V-shaped which provides on one leg of the V, a surface for mounting to the interior panel and the other leg a surface for applying the flocking which meets the window, and serves as a channel to catch liquids or debris from falling into the door cavity. The profile of a beltline window seal may have a plurality of such V shapes.

(13) In the general center of a car interior door panel is typically located a rigid door pull having a well on its top side which provides a hand hold for a passenger to insert their fingers to pull the door shut.

(14) The protective shell may be vacuum formed from plastic of sufficient gauge and rigidity to maintain its shape but which is also sufficiently deformable such that it can smoothly surmount interferences between itself and the vehicle during its installation without causing damage to the vehicle surfaces. A major portion of the shell matches closely to the surface of the interior car panel and therefore is made for a specific model of vehicle. When installed on the vehicle, the protective shell covers the majority of the interior panel including the window sill, the door pull which is sometimes integrated into the arm rest, and the upper plane of the panel where most of the soft or

highly finished surfaces of the door are located. The protective shell is cut out at the striker location so as not to interfere with the operation of the door closing. It can be optionally cut out to expose desired features such as the door opening handle, window buttons, door lock buttons, speakers and cup holders.

(15) The protective shell, following the shape of the car interior panel, may be generally planar, have a molded feature which protrudes toward the interior of the vehicle to cover the door pull and may include an integrally molded downward flange at its top edge for inserting into the V channel of the beltline interior window seal. The molded flange which extends the length of the V channel is sufficiently thin to insert into the V channel of the seal without impeding its sealing function. The protruding feature on the protective shell has a well on its top surface whose width follows the width of the well on the door pull to lock the shell into its service position. This locking well in the protective shell, working together with its major surface, which is generally planar and follows the contours of the door panel, serves to adequately lock the protector onto the door panel when it is installed in its service position while the return flange at its top edge serves to further secure it to the door panel. The well of the protective shell has a generous fillet where its walls meet its bottom surface to allow the shell to glide smoothly across the interior precipice of the door pull well during installation until it snaps into its service position. The locking well of the protective shell also provides the ability of the occupant to use it as a door pull.

(16) In many vehicles, the door pull generally has a bottom surface which is approx. 3" to 5" below its top surface. If the shell were to perfectly match the contour of the door pull, it would create an interference which would prevent the insertion of the well of the protective shell into the door pull well. In order to overcome this obstruction, the bottom surface of the shell's door pull protrusion is offset downward from its corresponding door pull surface by a distance which at least is equal to the depth of the locking well on its top surface to provide adequate clearance to fit the protrusion over the door pull. This offset creates a cavity underneath the door pull when the shell is fully installed.

(17) The protective shell is installed by initially setting it onto the door pull such that the bottom surface of its locking well sits on top of the door pull. Then, with light finger pressure applied downward and towards the door onto the upper edge of the shell's protrusion, the reverse side surfaces of the protective shell will glide smoothly across the panel at the precipice of the door pull well and across other slightly interfering points along the surface of the door until it snaps into position. Hence, the protective shell must be hard enough to be self-supporting but sufficiently pliable and resilient in order to facilitate its smooth installation and also to prevent damage to the delicate surfaces of the interior door panel during installation.

(18) Since almost all of the potential causes of damage will have either a downward or sideward trajectory, the protective shell remains securely in its service position to resist forces from these directions until it is ready to be removed.

(19) To unlock the protective shell, applying light finger pressure upward against the bottom surface of the door pull protrusion of the protective shell will unlock the locking well from the door pull which releases the shell.

(20) A suitable protective shell can be manufactured by vacuum forming a plastic sheet of impermeable plastic such as PET (polyethylene terephthalate) which is of sufficient thickness (approx. 0.060") to maintain its shape but appropriately pliable such that it can smoothly surpass minor interferences between itself and points on door panel when it is snapped into its service position on the interior door panel using only light finger force, yet sufficiently hard that it is resistant to abrasion and impact during use. The shell will spring back to its original shape once it surpasses the interference points keeping a snug fit against the door. The inner surface of the shell is smooth to prevent abrasion during installation and use. PET plastic is the most widely produced plastic in the world today and has many applications including its more recent use in the manufacture of hard sided luggage, examples of which are exposed to abrasion and impact from

other objects during handling. Other materials that may be used to create the protective shell are Polycarbonate, ABS (acrylonitrile butadiene styrene) or HIPS (high impact poly styrene).

(21) The flange of the protective shell located at the top end of the window sill needs to be sufficiently thin in order to fit inside the V channel of the window seal such that it does not impede its sealing purposes. Simultaneously, the remainder of the shell must be sufficiently thick to provide adequate protection for the door from damage. The vacuum forming process, which involves heating a parison sheet of thermoplastic until it is malleable, drawing it down onto a mold by manipulating relative air pressure, and allowing it to cool in its new shape until it is stiff, is well suited for the creation of plastic parts with thin profiles relative to their overall size. It is therefore ideal for forming the protective shell of the present invention since one characteristic of vacuum forming is that the parison sheet stretches as it is drawn down and more so where it is adjacent to a vertical wall becoming progressively thinner as it is drawn toward the bottom of the wall. Since the protective shell is vacuum formed laying flat on its reverse face, the flange will become the thinnest portion of the shell while its major planar surface, its smooth transition into the sill, and the protrusion that covers the door pull will be first to make contact with the parison, cooling it in these areas, and therefore locally stopping it from stretching further. The vacuum forming process allows the protective shell, inclusive of its components, to be formed as a single integral piece since it naturally forms the shell thicker in the areas that will be exposed to more abrasion and impact while forming the flange to be sufficiently thin. Forming the protective shell as a single piece simplifies the construction of the protective shell thereby lowering its cost to manufacture and enhancing its structural integrity.

(22) In some cases, undercut features such as the locking well described in the present invention may be difficult for certain vacuum form manufacturers to form. An alternative to forming the locking well and the main body of the shell in one shot is to form a locking well separately, which is then inserted into a receiving hole cut out on the door pull protrusion. This does not change the operation of the protective shell or its characteristics as described above. The installation may differ since the protective shell is first fitted directly onto the door panel and then the locking well is inserted into the hole cutout on the shell. Removal remains the same as above.

(23) In accordance with another aspect of the invention there is provided a vehicle door assembly, comprising: an interior door panel having an inner face with a protruding door pull defining a door well for receiving a user's fingers and a door sill leading to a window opening; and a protective shell for protecting the door panel against interior damage, the protective shell comprising: a deformable sheet of hard plastic with an exposed side and a reverse side, the reverse side being profiled to fit over the inner face of the vehicle door interior panel; an outwardly directed top surface portion for resting on the door sill; a protrusion on the exposed side of the deformable sheet for enclosing the vehicle door pull, the protrusion having an upper ledge; a locking well extending downwardly from the upper ledge snugly fitting into the door well to hold the protective shell securely in place on the interior door panel; and a separate flange component having a main surface and a downward flange at its upper extremity tuckable over a top edge of the door sill, said main surface being attachable to said outwardly directed top surface portion.

(24) In yet another aspect the invention provides a method of protecting a vehicle door comprising an interior door panel having an inner face with a protruding door pull defining a door well for receiving a user's fingers and a door sill leading to a window opening, the method comprising: tucking a flange component having a main surface and a downward flange at its upper extremity over a top edge of the door sill; applying a protective shell to the inner face of the interior door panel, the protective shell being in the form of a deformable sheet of hard plastic having a protrusion enclosing the door pull, such that the protrusion rests on the door pull; applying downward pressure to a locking well coupled to the protrusion to insert the locking well into the door well to form an interference fit retaining the protective shell in position against the interior

door panel; and attaching said main surface of said flange component to an outwardly directed top surface portion of said protective shell.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The invention will now be described in more detail, by way of example only, with reference to the accompanying drawings, in which:
- (2) FIG. 1 is a perspective view of a protective shell for a passenger car door interior panel showing the side exposed to the interior of the vehicle.
- (3) FIG. 2 is a perspective view of the protective shell of FIG. 1 showing the reverse side.
- (4) FIG. 3 is an exploded perspective view showing the protective shell of FIG. 1 aligned for mounting to the vehicle door interior panel.
- (5) FIG. 4 is a perspective view of the protective shell of FIG. 1 installed onto the door of FIG. 3.
- (6) FIG. 5 is a perspective view of the protective shell of FIG. 1 installed onto the door of FIG. 3.
- (7) FIG. 6 is a section view of upper portion FIG. 5 in elevation taken at the line 2-2 of FIG. 5
- (8) FIG. 7 is a section view of FIG. 5 in elevation taken at the line 2-2 of FIG. 5 showing the protective shell resting on the door pull of the vehicle interior door panel immediately before it is snapped into service position.
- (9) FIG. 8 is a section view of FIG. 5 in elevation taken at the line 2-2 of FIG. 5 showing the protective shell which is installed in its service position on the door panel.
- (10) FIG. 9 is a section view of FIG. 5 in elevation taken at the line 2-2 of FIG. 5 showing the protective shell immediately after it is unsnapped from its service position on the door interior panel.
- (11) FIG. 10 is an enlarged sectional detail view of FIG. 7 showing the inner door beltline window seal of the vehicle and the protective shell.
- (12) FIG. 11 is an enlarged sectional detail view of FIG. 8 showing the inner door beltline window seal of the vehicle and the protective shell.
- (13) FIG. 12 is a perspective view of a variation of a protective shell of FIG. 1 showing a door pull locking well which is manufactured as a separate part from the main body of the protective shell.
- (14) FIG. 13 is an enlarged section view of the protective shell of FIG. 12 in elevation taken at the line 3-3 showing the door pull locking well which is uninstalled.
- (15) FIG. 14 is a view of the protective shell of FIG. 13 in showing the door pull locking well installed in locked position onto the main body of the protective shell.
- (16) FIG. 15 is an enlarged detail view of FIG. 13 showing the door pull locking well being inserted into a receiving hole on the body of the protective shell of FIG. 12 and also showing the car door of FIG. 3.
- (17) FIG. 16 is an enlarged detail view of FIG. 14 showing the door pull locking well installed in locked position onto the main body of the protective shell.
- (18) FIG. 17 is a sectional view showing a parison and mold of a vacuum forming machine;
- (19) FIG. 18 is a sectional view shown the formed product on the mold;
- (20) FIGS. 19 to 23 illustrate an alternative embodiment of the separate door pull locking well;
- (21) FIGS. 24-26 show an embodiment of the protective shell shaped to make an interference fit with the vehicle interior door panel;
- (22) FIGS. 27-29 illustrate the locking well serving as a door pull;
- (23) FIGS. 30-32 show an alternate embodiment of the protective shell lacking a downward flange in combination with a separate flange component.

DETAILED DESCRIPTION OF THE DRAWINGS

- (24) Referring to FIG. 1 and FIG. 2, a protective shell 10 for a passenger car comprises an element

shaped to fit over the original equipment door interior and interior door panel of a passenger car. The protective shell has an exposed side **25** shown in FIG. **1** that remains exposed to the interior of the vehicle while it is installed on the vehicle and a reverse side **26** shown in FIG. **2**, which is mainly in contact with the original equipment door interior panel of the vehicle.

(25) The protective shell **10** defines a mainly planar shape **13**, a top surface **17**, a downward flange **12** at the extremity of top surface **17**, a protrusion **14** which extends toward the interior of the vehicle, and a locking well **11** located on top ledge **15** of the protrusion **14**. In the example shown, the transition between the main planar shape and the top surface **17** is smoothly rounded.

(26) As shown in FIG. **3**, the protective shell **10** is positioned onto the door **50** interior of the vehicle such that the flange **12** tucks over the top edge **58** of the interior door panel **52**, the top surface **17** rests on top of interior door panel sill **57**, the planar surface **26** lays mainly against the major face **63** of the door interior panel **52** and the protrusion **14** extends from the major plane of the protective shell **10** encloses the vehicle door pull **53**. The locking well **11** is tucked into a roughly matching well **54** located on the top side of the door pull **53** of the interior door panel **52**. FIG. **4** shows the protective shell **10** correctly positioned while it is installed on the vehicle door **50** interior.

(27) As best shown in FIG. **4**, the protective shell **10** is sized and shaped to cover a vehicle's original equipment interior door panel **52** and portions of the interior door **50** that are most often exposed to potential damage from its occupants or cargo, such as the vehicle's interior window sill **57**, the major plane of the interior panel **63** and door pull **53**, and especially the top surface of the door pull **55** which, in many modern vehicles, also serves as an armrest. The protective shell **10** is cut out at **24** to allow clearance for the striker on the vehicle corresponding pillar. Other portions of the protective shell **10** may optionally be cut out to expose or allow access to speakers, storage, door opening handle, door controls and window controls. The door example **50** in FIG. **3** is the left side rear door of a 2019 Jeep Wrangler JL.

(28) As shown in FIG. **7**, to install the protective shell **10** onto the vehicle door **50**, it is first positioned with its reverse side **26** against the major plane of the interior door panel **63**, being temporarily supported by the bottom face **27** of the locking well **11** resting on the top surface of the door pull **55**. Because the bottom surface **16** of the protective shell **10** is sufficiently offset by at least the depth of the locking well **11**, the protrusion **14** will fit over the door pull **53**.

(29) Using light finger pressure, it is then pushed at the center of the top edge **20** of the protrusion **14** downward and toward the interior door panel at which point, the bottom surface **27** of the well **11** will slide over the precipice **59** of the door pull well **54**. Because most car door interior panels **52** are designed with various features and contours, points on the panel **52** such as **62** and **63** will inherently interfere with corresponding surfaces **18** and **19** of the protective shell and provide slight resistance. However, as the shell **10** is appropriately deformable due to the proper selection of material and gauge, it will generally buckle and allow the bottom fillet **23** of the well **11** to glide smoothly over the precipice **59** of the door pull **53** until the well **11** of the protective shell **10** snaps securely inside the well **54** of the door pull **52**. Because the reverse surface **26** and flange **12** of the protective shell **10** are closely matched to the door panel **52**, during the same motion, the flange **12** will automatically seat itself into the correct position inside the V channel **72** of the beltline interior window door seal **70** (FIG. **10** and FIG. **11**) thus, preventing contact with the delicate fibers of the flocked side **71** of the of the seal. No part of the shell **10** will make contact with the window **51** once it is installed thus allowing the window **51** to be rolled up or down freely without causing damage to its surface **65**. The walls **22** of the well **11** of the protective shell **10** which roughly match the interior walls **60** of the door pull well **54** will secure the shell in position until it is ready to be removed. The flange **12** and various interfering features such as **62** and **63** will provide additional locking support to keep the shell **10** in place.

(30) To uninstall the protective **10** shell from the door panel **52**, light finger pressure is applied upward on the bottom surface **16** of the protrusion **14** as shown in FIG. **9**. This action will lift the

entire shell **10** thereby unseating the well **11** of out of its seated position within the well **54** of the door pull **53** and release the protective shell **10**.

(31) In an alternate embodiment of the invention, the locking well **102** of the protective shell is described in FIG. **12** as a separate piece which is inserted into a receiving hole cut out **103** on the door pull protrusion **14**. FIG. **13-16** are section views of 3-3 of FIG. **12**. The well **102** is inserted into the hole **103** of the protective shell **101** and therein locked into position by the engagement of the cut edge **106** of the shell **101** and a small groove **105** extending around the perimeter of the well **102** just below its top edge. FIG. **15** shows an alternate method of installation where the protective shell **101** is placed over the interior panel **52** and the well **102** is inserted in the hole **103** afterward. In this case, the wall **104** of the well **102** which is locked into the shell **101** will serve the same function as the wall **22** of FIG. **6** to resist movement of the shell **101** away from the surface **63** of the interior panel **52** until it is ready to be removed. Therefore, this alternate embodiment does not change the function of the protective shell **10** or its properties as described above, even though it is manufactured as a perforated body **101** and a separate well **102**.

(32) As shown in FIGS. **17** and **18** the parison is vacuum formed over the mold **80**. It will be noted that the fillet **27** at the base of the well **11** and the flange **12** are thinned as a result of the vacuum forming process. In one exemplary embodiment the thickness can vary from 0.06" in the major plane of the shell and the vertical face of the door pull protrusion to 0.03" for the fillet **27** at the bottom of the locking well **11** and the flange **12**. This helps to enhance the resilience of the locking well **11** to facilitate locking in the matching well **54** of door pull and reduce the risk of the flange **12** damaging the window **51**.

(33) It will be noted that the described protective shell is removably securable to the vehicle door panel without need for fasteners or adhesives by the interface with the well of the door pull handhold and a corresponding well of the panel, is self supporting, and through the combination of its thickness and material properties is sufficiently pliable and resilient to surpass slight interferences against points on the door during installation.

(34) The protective shell **10** may be installed and removed frequently which over time may cause fatigue on the locking well **102**. Therefore, a stronger assembly may be needed to extend the longevity of the protective shell **10**. FIGS. **19-22**, show an alternate embodiment to the separate door pull locking well which is a two-part locking well assembly **102** where there is an inner well component **102a** and an outer well component **102b** that comprises the locking well **102**. In one embodiment the two components are formed from rigid thermoplastic such as ABS of approximately 0.075" in thickness.

(35) The inner well component **102a** of the locking well **102** is a cupped shape that extends vertically and terminates at an outwardly protruding flange around its upper perimeter. The flange of the inner well **102a** is generally perpendicular to the vertical wall of the inner well **102a** and the underside of the flange matches the top surface of the top ledge **15** of the protrusion **14** of the shell **101**. At the level of the flange, the outer surface of the vertical wall of the inner well component **102a** exactly matches the perimeter of the receiving hole cutout **103** of the shell **101** such that the inner well component **102a** can be inserted tightly into the hole cutout **103** from the top of the ledge **15** of the protrusion **14** until it is stopped by the perimeter flange. The outer component **102b** of the two-part locking well **102** is also a cupped shape that extends vertically and whose inner surface closely matches the outer surface of the inner well component **102a** and terminates at a perimeter lip at its upper end which matches the reverse surface of the protective shell **101** at the top ledge **15** of the protrusion **14**. FIG. **22** The outer well component **102b** is then placed over the bottom of the of the inner well component **102a** and fixed together by means of screws, snaps or glue such that they will pinch top ledge **15** of the shell **101** between the flange of the inner well component **102a** and the top perimeter lip of the outer well component **102b** at the receiving hole cutout **103** as shown in FIGS. **20-22**.

(36) In the embodiment shown in FIGS. **24-26** the protective shell **10** is shaped such that areas of

its perimeter at the front, rear and bottom are in interference with the vehicle interior door panel **52** when it is installed on the door **50**. Because the shell **10** is securely held in position on the door **50** by the locking well **102** and the shell **10** is appropriately deformable due to the proper selection of material and gauge, it will generally buckle at the said areas of interference around its perimeter to create a pre-sprung condition which enhances the snug fit of the shell **10** against the door **50**.

(37) Because the locking well **102** is cup shaped, and is inserted securely inside the well **54** of the door pull **53**, the locking well **102** will allow a passenger to insert their fingers into its cavity well as illustrated in FIGS. **27-29**. Thus, the locking well **102** of the protective shell **10** provides the ability of the occupant to use it as a door pull.

(38) Since the gap between the window **51** and the top edge **58** on some vehicles is very tight, the integrated flange **12** of the protective shell **10** may not easily self-locate into the intended position over the top edge **58** of the door panel during installation. FIGS. **30 to 32** show an alternate embodiment of the protective shell **201** which is absent of a downward flange **12** at its upper extremity, and a separate flange component **212**. The separate flange component **212** is comprised of a main surface which matches top surface of the interior door panel sill **57** and turns sharply into a downward flange at its upper extremity. A strip of double-sided pressure sensitive adhesive **212a** is affixed to the top of the separate flange component **212**. The separate flange component **212** is first positioned on top the interior door panel sill **57** such that the downward flange is tucked over top of the edge **58**. The protective shell **201** is then installed by inserting the locking well **102** into the roughly matching well **54** of the vehicle door interior panel **52** which will cause the protective shell **201** to self-locate onto the interior door panel **52** such that the top surface **17** of the protective shell will fall onto the installed separate flange component **212**. Light finger pressure on the top surface **17** of the protective shell **201** will activate the double-sided pressure-sensitive adhesive **212a** and adhere the separate flange component **212** with to the protective shell **201** and further secure it to the interior door panel **201**. The strip of pressure-sensitive **212a** on top of the separate flange component **212** may be replaced by temporary fasteners such as hook and loop fabric such as Velcro, or button snaps to make the assembly reusable and the installation process repeatable. The function and installation process will be the same as the pressure-sensitive adhesive and the removal of the protective shell will be performed by pressing upward on the underside of the protrusion **14** to disengage the locking well **102** and the downward flange of separate flange component **212**.

Claims

1. A protective shell for protecting an interior door panel of a vehicle, the interior door panel having an inner face with a protruding door pull defining a door well for receiving a user's fingers and a door sill leading to a window opening, the protective shell comprising: a deformable sheet of hard plastic with an exposed side and a reverse side, the reverse side being profiled to fit over the inner face of the vehicle door interior panel; an outwardly directed top surface portion for resting on the door sill; a protrusion on the exposed side of the deformable sheet for enclosing the vehicle door pull, the protrusion having an upper ledge; a locking well extending downwardly from the upper ledge and responsive to downward pressure to fit snugly into the door well to hold the protective shell securely in position on the interior door panel; and a separate flange component having a main surface and a downward flange at its upper extremity tuckable over a top edge of the door sill, said main surface being attachable to said outwardly directed top surface portion.
2. The protective shell of claim 1, wherein said main surface is coated with a pressure-sensitive adhesive for attachment to said outwardly directed top surface portion.
3. The protective shell of claim 1, constructed of a resilient material and configured to form a pre-sprung interference fit with said interior door panel.
4. The protective shell of claim 1, wherein said locking well is a separate two-part assembly fixable

in a matching cutout in said upper ledge and comprises an inner well component and an outer well component; and said inner locking well component has a peripheral flange overlapping a perimeter of said matching cutout, and said outer well component fits snugly over a bottom portion of said inner well component to pinch the upper ledge between said peripheral flange and a perimeter lip of said outer well component.

5. The protective shell of claim 4, wherein the inner locking well component snaps into said outer locking well component.

6. The protective shell of claim 5, wherein the two-part assembly is made of a rigid thermoplastic.

7. The protective shell of claim 4, wherein the locking well is sized to permit entry of a person's fingers to allow said locking well to be used as a door pull.

8. A vehicle door assembly, comprising: an interior door panel having an inner face with a protruding door pull defining a door well for receiving a user's fingers and a door sill leading to a window opening; and a protective shell for protecting the door panel against interior damage, the protective shell comprising: a deformable sheet of hard plastic with an exposed side and a reverse side, the reverse side being profiled to fit over the inner face of the vehicle door interior panel; an outwardly directed top surface portion for resting on the door sill; a protrusion on the exposed side of the deformable sheet for enclosing the vehicle door pull, the protrusion having an upper ledge; a locking well extending downwardly from the upper ledge snugly fitting into the door well to hold the protective shell securely in place on the interior door panel; and a separate flange component having a main surface and a downward flange at its upper extremity tuckable over a top edge of the door sill, said main surface being attachable to said outwardly directed top surface portion.

9. The vehicle door assembly of claim 8, wherein said main surface is coated with a pressure-sensitive adhesive for attachment to said outwardly directed top surface portion.

10. The vehicle door assembly of claim 8, which is constructed of a resilient material that forms a pre-sprung interference fit with said interior door panel.

11. The vehicle door assembly of claim 8, wherein said locking well is a separate two-part assembly fixable in a matching cutout in said upper ledge and comprises an inner well component and an outer well component; and wherein said inner locking well component has a peripheral flange overlapping the perimeter of said matching cutout, and said outer well component fits snugly over a bottom portion of said inner well component to pinch the upper ledge between said peripheral flange and a perimeter lip of said outer well component.

12. The vehicle door assembly of claim 11, wherein the locking well is sized to permit entry of a person's fingers to allow said locking well to be used as a door pull.

13. The vehicle door assembly of claim 11, wherein the two-part assembly is made of a rigid thermoplastic.

14. A method of protecting a vehicle door comprising an interior door panel having an inner face with a protruding door pull defining a door well for receiving a user's fingers and a door sill leading to a window opening, the method comprising: tucking a flange component having a main surface and a downward flange at its upper extremity over a top edge of the door sill; applying a protective shell to the inner face of the interior door panel, the protective shell being in the form of a deformable sheet of hard plastic having a protrusion enclosing the door pull such that the protrusion rests on the door pull; applying downward pressure to a locking well coupled to the protrusion to insert the locking well into the door well to form an interference fit retaining the protective shell in position against the interior door panel; and attaching said main surface of said flange component to an outwardly directed top surface portion of said protective shell.

15. The method of claim 14, wherein said main surface of said flange component is attached to said surface portion of said protection shell by means of a pressure-sensitive adhesive.

16. The method of claim 14, wherein the protective shell is made of a pre-sprung resilient material and is applied to said interior door panel by making an interference fit with said interior door panel.

17. The method of claim 14, wherein the protrusion has an upper ledge for resting on the door pull,

said upper ledge being formed with a cutout; and wherein prior to applying the protective shell to the inner face of the interior door panel an inner locking well component with a peripheral flange is inserted through said cutout into an outer well component such that said inner locking well component and said outer locking well component pinch said upper ledge between a perimeter lip of said outer locking well component and said peripheral flange to form said locking well.

18. The method of claim 17 wherein said inner locking well component is snapped into said outer locking well component.
